

LAB 11

NAME - AADITYA AMLAN PANDA

ROLL:220007

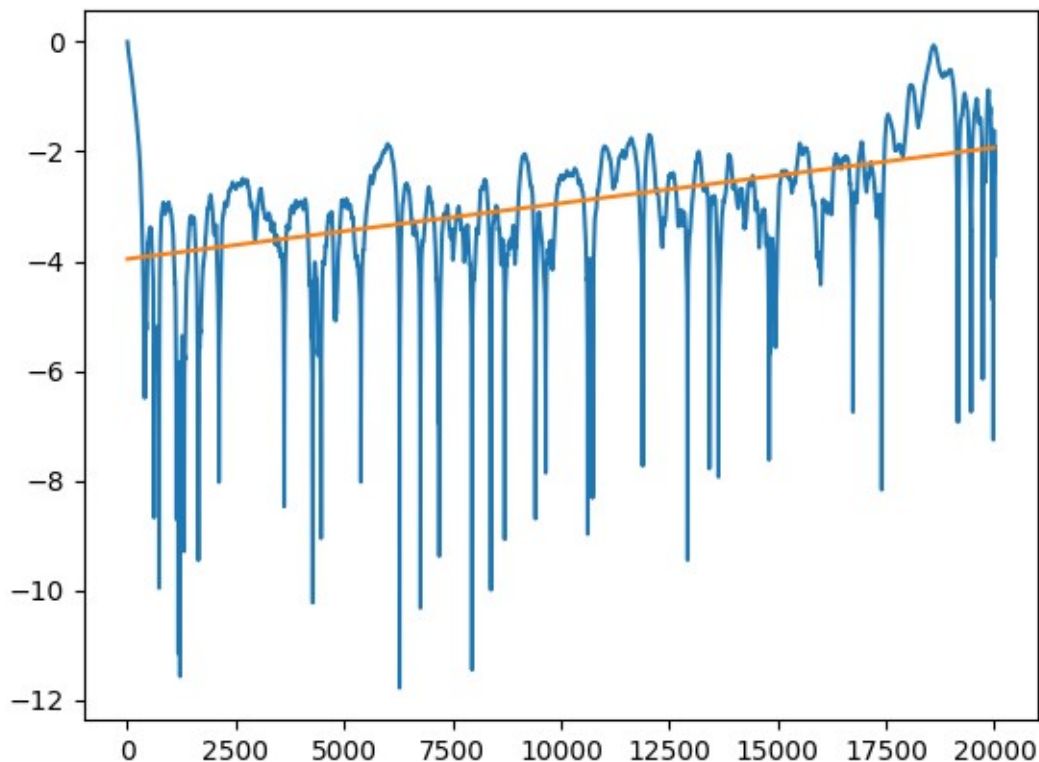
Methodology:-

- **Question 1:**
 - First the correlations were calculated at each lags.
 - Log of the correlations were fitted with k (timepoints) using Ordinary Least squares
 - This was repeated only for the first 500 points(to avoid fluctuations) to obtain relaxation time using $\tau = -1/m$
 - The model was determined to Autoregressive due to non zero ACF and the order was calculated observing when sign of PACF changes.
- **Question 2:**
 - The Spencer's 15 point strategy was used to smoothen the curve the weights used were $1/320 \times [74, 67, 46, 21, 3, -5, -6, -3]$
 - The curves were plotted for first 500 points. First 7 points were omitted
 - The smooth curve was fitted and the fitted values were subtracted from the true values to obtain the detrended plot.

Results and Observations:-

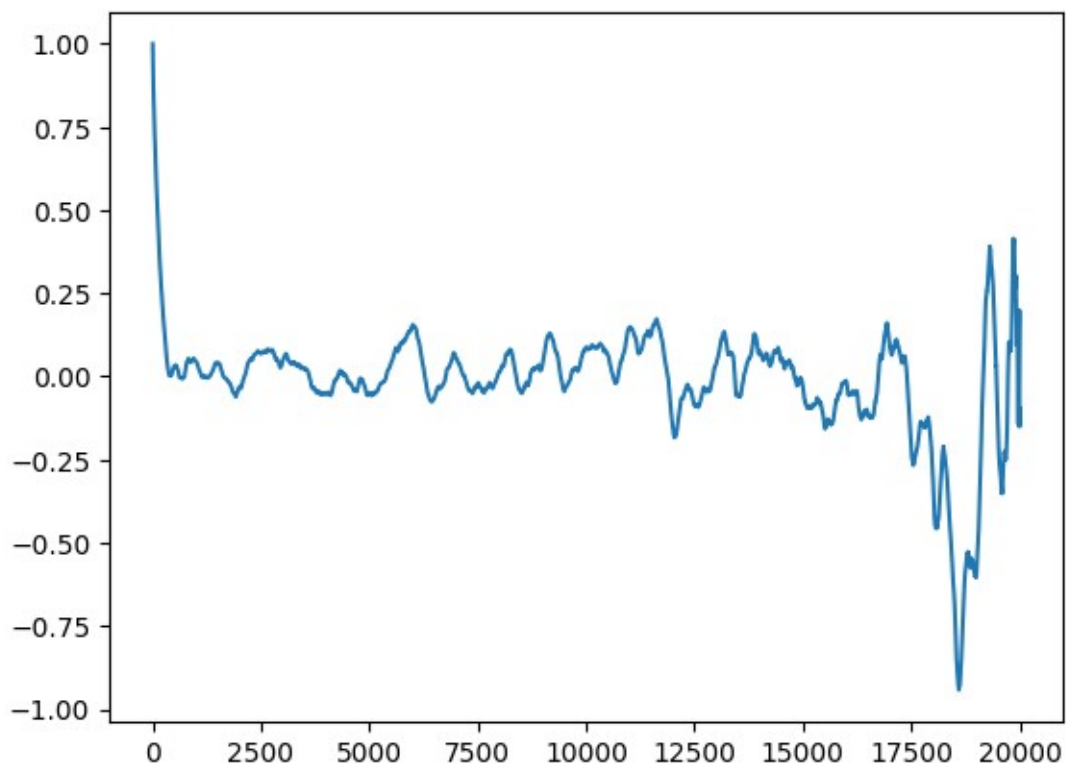
Question1: -

t vs log (rho) plot: -



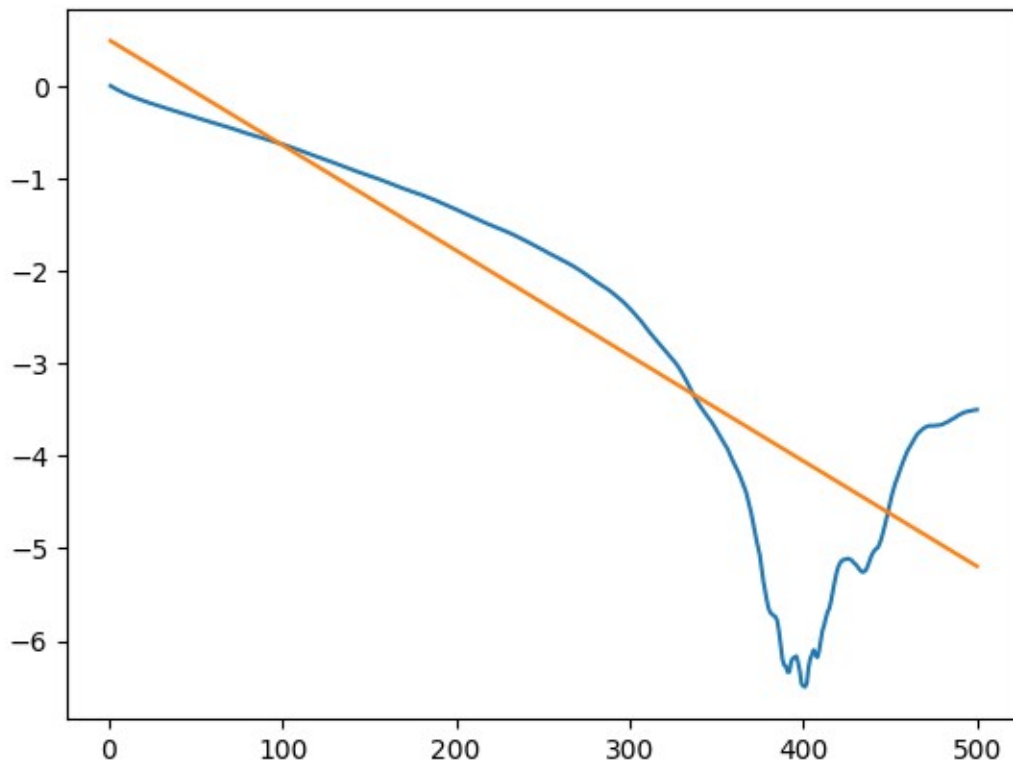
$m = 0.00010154956130041927$
 $c = -3.9597920295767572$

t vs log (rho) plot: -



ACF never becomes 0 in the above plot. Hence not MA

t vs log (rho) plot (first 500 points): -



$m = -0.011378386556363078$

$c = 0.4965296150354316$

$\tau = 87.88592258194771 = \text{RELAXATION TIME}$

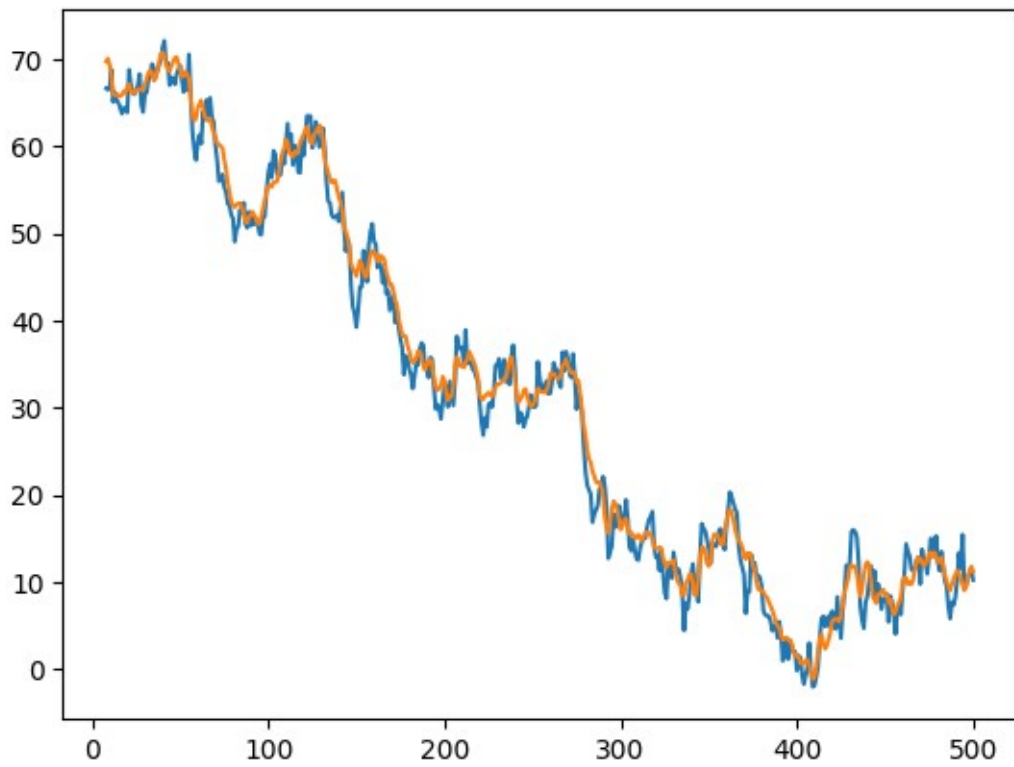
PACF values: -

[1. 0.999 -0.043 -0.03 -0.021 -0.007 -0.003 -0.008 -0.009 -0.011
 -0.015 -0.014 -0.015 -0.011 -0.004 -0.004 -0.004 -0.004 -0.008 -0.005
 -0.005 -0.006 -0.004 -0.004 -0.003 -0.002 -0.004 -0.008 -0.01 -0.014
 -0.016 -0.012 -0.014 -0.014 -0.012 -0.013 -0.013 -0.013 -0.012 -0.01
 -0.007 -0.005 -0.004 -0.004]

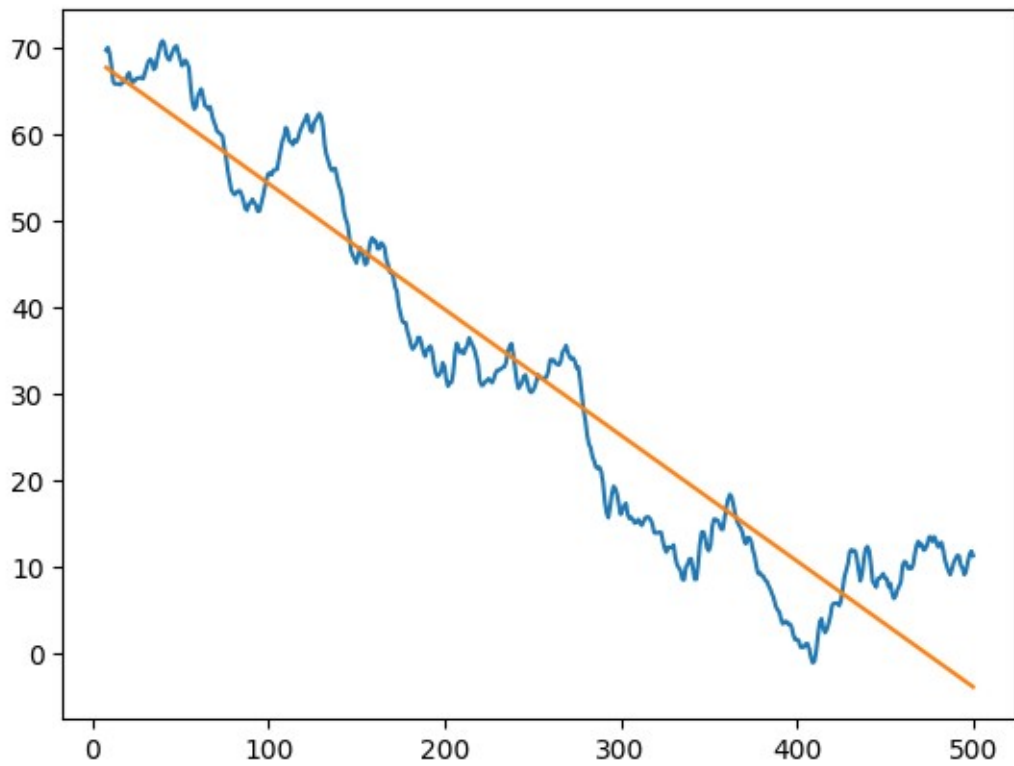
It can be observed that the sign changes after second hence the model is an **AR(2)** model

Question 2:

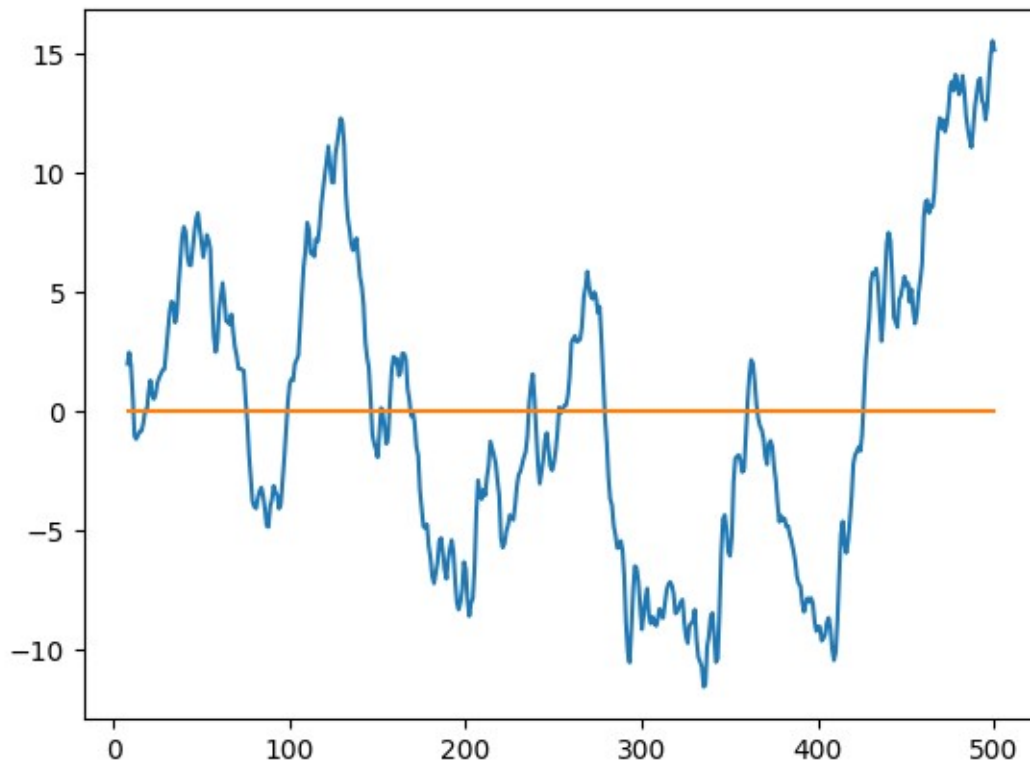
t vs value plot for true and smoothened data: -



t vs value plot (fitting smoothened data): -



t vs value plot (detrended) :-



Appendix:-

Question 1:-

```
import numpy as np
import pandas as pd
import math
import matplotlib.pyplot as plt

np.set_printoptions(precision = 3, suppress = True)

def find_Ck(x,y,z,k):
    sum = 0.0
    N = len(x)
    for i in range(N):
        if i + k >= N:
            break;
        sum += (x[i]*x[i+k] + y[i]*y[i+k] + z[i]*z[i+k])
    return sum/(N-k)

def fit_linear(x,y):
    sum_xy = 0.0
    sum_x = 0.0
    sum_y = 0.0
    sum_x2 = 0.0
    n = len(x)
    for i in range(n):
        sum_xy += x[i]*y[i]
```

```

sum_x += x[i]
sum_y += y[i]
sum_x2 += x[i]*x[i]
m = (n*sum_xy - sum_x*sum_y)/(n*sum_x2 - sum_x**2)
c = (sum_y - m*sum_x)/n

return m,c

data = pd.read_csv('/content/lab11_polymer.csv').values
t = data[:,0]
x = data[:,1]
y = data[:,2]
z = data[:,3]
N = len(t)

C = np.zeros(N)
logrho = np.zeros(N)
rho = np.zeros(N)

for i in range(N):
    C[i] = find_Ck(x,y,z,i)
    rho[i] = C[i]/C[0]
    logrho[i] = math.log(abs(rho[i]))

m,c = fit_linear(t,logrho)

plt.plot(t,logrho)
plt.plot(t,m*t+c)
plt.show()

data = pd.read_csv('/content/lab11_polymer.csv').values
t = data[:,0]
x = data[:,1]
y = data[:,2]
z = data[:,3]
N = len(t)

m,c = fit_linear(t,logrho)

plt.plot(t,logrho)
plt.plot(t,m*t+c)
plt.show()
print(m)
print(c)

plt.plot(t,rho)

plt.show()

lim = 500

```

```

mt,ct = fit_linear(t[:lim],logrho[:lim])
plt.plot(t[:lim],logrho[:lim])
plt.plot(t[:lim],mt*t[:lim]+ct)
plt.show()
print(mt)
print(ct)
tau = -1/mt
print(tau)

from statsmodels.graphics.tsaplots import pacf, plot_pacf

pacf_values = pacf(rho,method='ywadjusted')

print(pacf_values)

```

Question 2:-

```

datag = pd.read_csv('/content/lab11_polymer.csv').values
t = datag[:,0]
xg = datag[:,1]

def spenc(X):
    w = [74,67,46,21,3,-5,-6,-3]
    w = np.array(w)/320
    Y = np.zeros(len(X))
    for i in range(len(X)):
        if (i+7)>=len(X):
            break
        for a in range(14):
            Y[i]+=w[a-7]*X[i+a-7]

    return Y

xsm = spenc(xg)

lim2 = 500
plt.plot(t[7:lim2],xg[7:lim2])
plt.plot(t[7:lim2],xsm[7:lim2])
plt.show()

mg,cg = fit_linear(t[7:lim2],xsm[7:lim2])
plt.plot(t[7:lim2],xsm[7:lim2])
plt.plot(t[7:lim2],mg*t[7:lim2]+cg)
plt.show()

plt.plot(t[7:lim2],xsm[7:lim2]-mg*t[7:lim2]-cg)
plt.plot(t[7:lim2],np.zeros(lim2-7))
plt.show()

```